MATH4840 Final Paper

Applications of Representation Theory
to Quantum Machine Learning

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1 Introduction

In using representation theory to solve GQML problems, our aim is to make it easier to identify exploitable symmetries in our data, and exploit them in the design of our model. In doing so, we hope to outperform standard "symmetry-agnostic" QML models.

In prior discussions of this subject we have discussed the topic of invariance of GQML data under group actions, and the topic of basic group representations. Some notions are recalled here:

1.1 Quantum Machine Learning

In quantum machine learning, instead of bits our data is composed of qubits, each of which is represented by a complex vector. An n qubit state, composed of n qubits, then has dimension 2^n. Given some data domain $\mathcal{R}$ in Hilbert space $\mathcal{H}$ with dimension 2^n, and some labels in $\mathcal{Y}$, our mission is to come up with a $h_\theta : \mathcal{R} \to \mathcal{Y}$ that accurately labels the data. In other words, if the data has some underlying map f that labels it, we want h_θ to be a close approximation of f. θ will be the set of parameters adjusted by our algorithm to achieve this approximation. The subject of this paper is not these parameters themselves, but the information we can glean about our data to inform our choice of θ.

1.2 Our formula and basic representation theory

We have been given a formula for h_θ, exemplary of a quantum machine learning model. We start with a finite data set $\rho = \{\rho_1, \rho_2, ..., \rho_n\}$, our n-qubit states, and then consider one n-qubit state ρ_i. Our h_θ is comprised of a neural network $\mathcal{W}_\theta$, a measurement operator M, or M_i, and $\rho_i^{\otimes k}$, the tensor product of k copies of ρ_i.

The formula itself we finally get is:

$$h_\theta(\rho_i) = \mathrm{Tr}[\mathcal{W}_\theta(\rho_i^{\otimes k})M_i]$$

Definition: We call a map equivariant under the action of some group if its domain and codomain are acted on by that group, and the action of that group commutes with the function. In other words, if $g \cdot f(x) = f(g \cdot x)$ for all $g \in G$, then f is equivariant under the group action of g.

Fact: It has additionally been proved in prior discussions that if $\mathcal{W}_\theta$ and M_i are equivariant under the group operation $\rho \to U\rho U^\dagger$ for any U in group G, that is, by definition

$$\mathcal{W}_\theta(U^{\otimes k}\rho_i^{\otimes k}(U^\dagger)^{\otimes k}) = U^{\otimes k'}\mathcal{W}_\theta(\rho_i^{\otimes k})(U^\dagger)^{\otimes k'}$$

and

$$U^{\otimes k}M = MU^{\otimes k},$$

then it is also true that

$$h_\theta(U\rho U^\dagger) = h_\theta(\rho)$$

In order to be able to transform our underlying data without affecting our labels, we will only consider G such that $\mathcal{W}_\theta$ and M_i are equivariant with respect to G.

The final subject we have discussed already is the concept of group representation.

Definition: A representation of group G acting on V is an $R : G \to GL(V)$. We will be primarily concerned with the case that $V = \mathbb{C}^2$ for some $k \in \mathbb{N}$, and $G = SU(2, \mathbb{C})$.

1.3 New background

Definition: An irreducible representation is a nonzero representation with no nontrivial invariant subspace (that is, no invariant subspace other than 0 and the whole space) under the group action.

Definition: For representation R of G on V, V is completely reducible if $V = W_1 \oplus W_2 \oplus ... \oplus W_k$, for subspaces $W_1, W_2, ..., W_k$ satisfying that W_i is invariant under the group action of $R|_{W_i}$ for all $i \in 1, ..., k$. Here $R|_{W_i}$ denotes the restriction of R to W_i, that is the elements r of R such that the entries of r are all in W_i. An R mapping onto a completely reducible V is also called completely reducible.

Note: We can visualize this more intuitively by noting that if R is completely reducible, there exists a basis in V in which

$$R(g) = \begin{pmatrix} R_1(g) & & \\ & \ddots & \\ & & R_k(g) \end{pmatrix}$$

for all $g \in G$. This block diagonal form will recur later.

Theorem 1: Any finite dimensional unitary representation is completely reducible.

Theorem 2: Any unitary operator has a unitary representation.

These theorems are given without proof, but will become quite important later.

2 A Version of Schur-Weyl Duality

2.1 Statement and proof

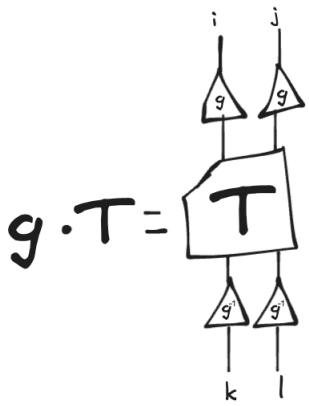
Theorem 3 (Schur-Weyl Duality): If there exist N distinct subspaces $U_1, U_2, ..., U_N$ of $V = \mathbb{C}^{n \otimes k}$ invariant under the group action of some representation $R : G \to GL(n, V)$ on V, there is a one to one correspondence between these subspaces and those arising from the group action of S_k on V. Furthermore, these group actions commute.

Here we are primarily concerned with the $V = \mathbb{C}^{2 \otimes k}$ and $G = SU(2, \mathbb{C})$ case, and the correspondence of these subspaces, so this is the case and fact we will prove.

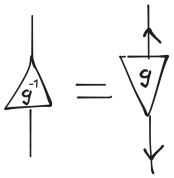
Proof: (Pictorial sketch)

We begin by considering some $T \in \mathbb{C}^{2 \otimes 2}$ which is invariant under the group action corresponding to $G = SU(2, \mathbb{C})$, so for any $g \in R(G)$, $g \cdot T = gTg^{-1} = T$. Pictorially, we can represent T as such:

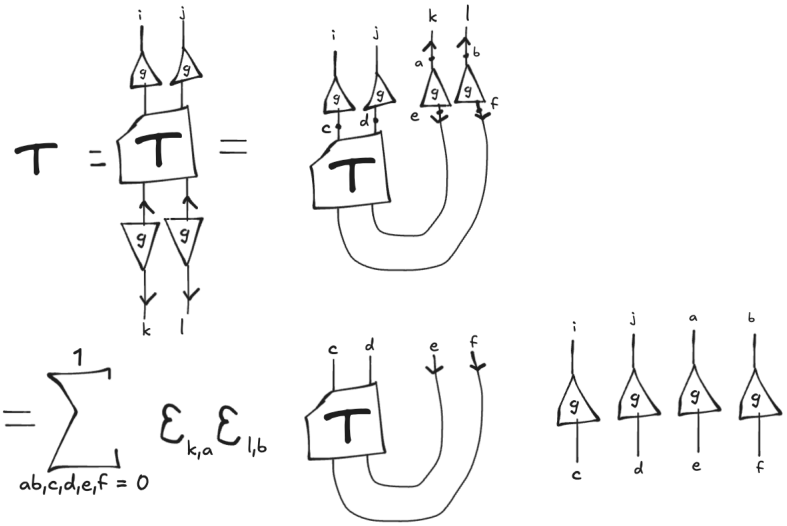
And:$g \cdot T$ as such:



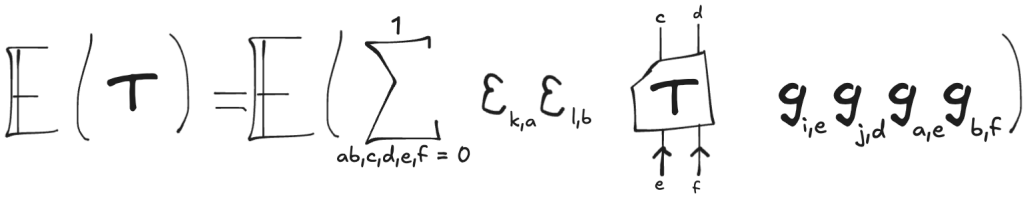
We can simplify this by employing an identity about these diagrams:

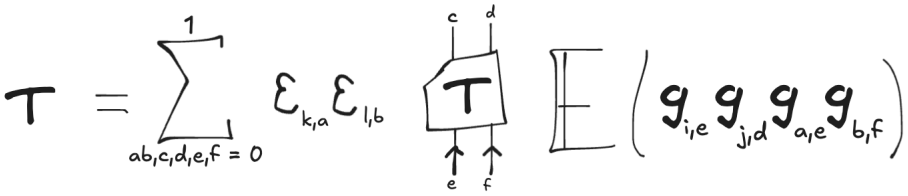


With this identity we get



From here we make another observation. Since T is a constant in $\mathsf{C}^{2\otimes 2}$, its expectation should just be $= T$, while the expectation of our right hand side can be simplified.





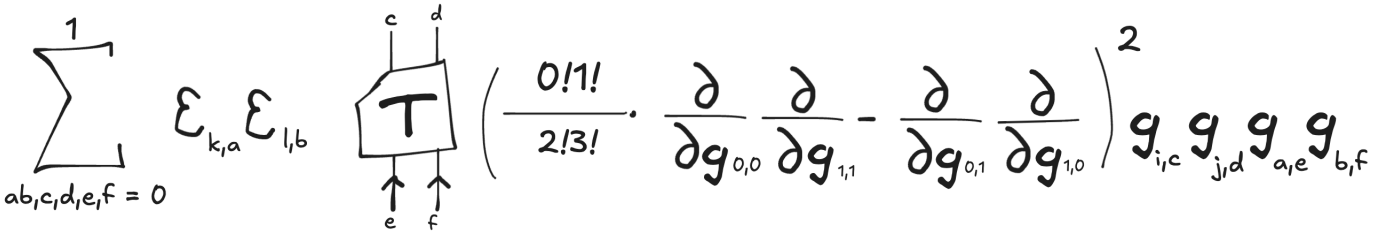
Now we apply another identity.

Lemma: For $k = 2l \geq 0$, $l \in \mathbb{N}$, $\quad I = \{i_1, i_2, ..., i_k\} \in [2]^k$, $J = \{j_1, j_2, ..., j_k\} \in [2]^k$, let $R_k(I, J) = \mathbb{E}_{SU(2,\mathbb{C}),\mathbb{C}}(f)$ where $f : SU(2, \mathbb{C}) \to \mathbb{C}$,

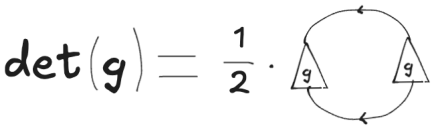
$g = \begin{matrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{matrix} \to g_{i_1 j_1} g_{i_2 j_2} ... g_{i_k j_k}$. Then it is also true that

$R_k(I, J) = \frac{0!!!}{l!(l+1)!}(\frac{\partial}{\partial g_{11}} \frac{\partial}{\partial g_{22}} - \frac{\partial}{\partial g_{12}} \frac{\partial}{\partial g_{21}})^l x_{i_1 j_1} x_{i_2 j_1]2} ... x_{i_k j_k}$.

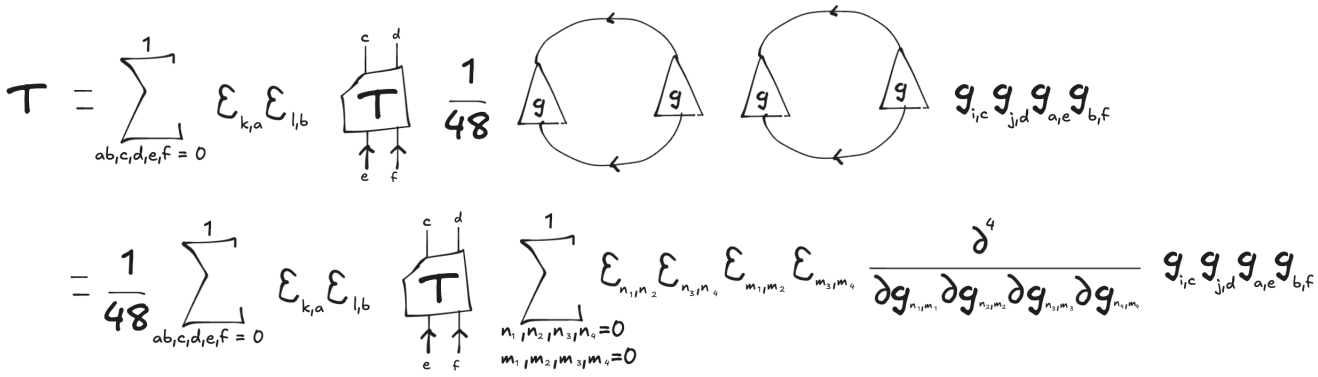
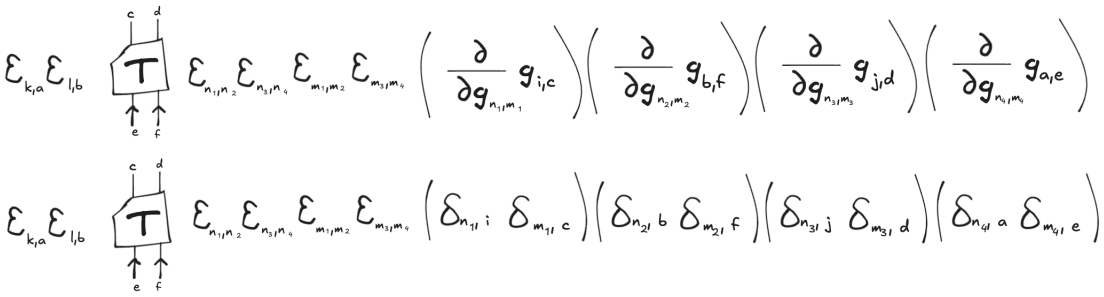
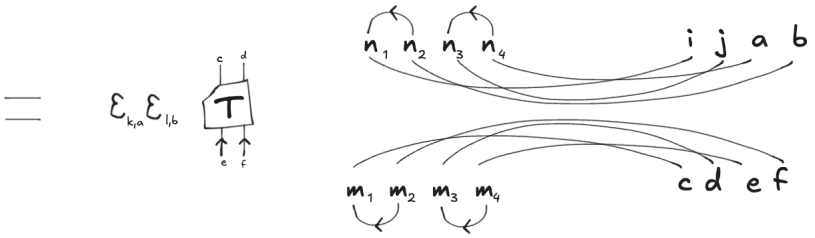
With this identity we get:



Again, we will apply one final identity for pictorial proofs:



and now our final expression to reduce is



Instead of solving for the result of this sum, we note that if we can find the form of any general term, then T will be a linear sum of such terms. Looking at this sum, we can see that each term will correspond to a set of assignments of n_1, n_2, n_3 and n_4 to i, j, a and b. Observation of the expression will show that the assignments of m_1, m_2, m_3 and m_4 to c, d, e and f will be identical. With that in mind, here we will compute just one term, corresponding to the element $(243) \in S_4$.